SPEFormer: A Synergistic Perception-Enhanced Transformer With Density-Guided Dynamic Queries for Remote Sensing Small-Object Detection

Ye Yuan, Shuanglong Li, Jian Liu, Zhaoshuo Diao, and Guoyu Tong

***Abstract*—Small-object detection in remote sensing images is highly challenging due to their limited pixel representation, weak feature expression, and strong interference from complex backgrounds. Moreover, existing query-based detectors typically use a fixed number of object queries for all images, making them poorly adaptive to sparse or dense target distributions, which may lead to missed detections or redundant computations. To address these issues, we propose SPEFormer, a detection framework specifically designed for small objects in remote sensing images. SPEFormer incorporates the selective global–local perception network (SGLPNet), which combines conditional convolution, global branches, and local branches across different stages to extract high-quality multiscale features, and the adaptive synergistic attention fusion (ASAF) module to adaptively fuse cross-level features, effectively enhancing small-object responses while suppressing background interference. In addition, a density-guided dynamic query (DGDQ) mechanism dynamically allocates the number and positions of object queries based on multiscale density maps, enabling the model to handle both sparse and dense target scenarios. Experimental results on VisDrone2019, DIOR, and DOTA-v1.0 show that SPEFormer achieves competitive detection performance. In particular, SPEFormer obtains strong AP_{50} scores of 54.7%, 87.7%, and 65.3% on the three datasets. Code is available at: https://github.com/SLBY8/SPEFormer**

***Index Terms*—Dynamic queries, remote sensing images, small-object detection, Transformer.**

I. Introduction

OBJECT detection is a fundamental task in remote sensing image analysis and supports a wide range of practical applications, such as urban planning, traffic monitoring, maritime surveillance, disaster assessment, and military reconnaissance. Beyond fully supervised detection, weakly supervised object detection has also been explored to improve label efficiency by learning object localization from image-level annotations [1]. Benefiting from the rapid development of remote sensing technology, modern remote sensing images contain abundant spatial details and semantic information [2]. In addition, remote sensing image interpretation has also benefited from related image processing and analysis techniques, such as hyperspectral image super-resolution, hyperspectral unmixing, hyperspectral image compression, sample augmentation for land-cover classification, and SAR–optical multimodal image registration and fusion. These techniques aim to enhance image quality, analyze spatial–spectral characteristics, reduce the storage and computational burden caused by high-dimensional spectral bands, improve classification performance under limited annotated samples, or exploit complementary information from multimodal observations, thereby supporting downstream interpretation tasks [3], [4], [5], [6], [7]. For example, recent SAR–optical studies improve multimodal image registration by incorporating radar imaging priors, and enhance cloud and shadow removal by fusing shared and modality-specific features from SAR and optical images [8], [9]. Nevertheless, accurate object detection in such images remains challenging. Compared with object detection in natural scenes, remote sensing object detection is characterized by more severe scale variations, complex background interference, and arbitrary spatial distributions. Recent studies on maritime ship detection further show that multiscale variations, occlusions, indistinct boundaries, and background clutter remain major factors that cause missed detections and false alarms in optical remote sensing imagery [10]. Similarly, aerial person detection in drone-assisted search-and-rescue scenarios also suffers from sparse distributions, small-object sizes, weak visual responses, and complex imaging conditions, which further highlight the difficulty of robust small-object detection in aerial images [11]. Moreover, edge-oriented APD methods further indicate that practical aerial detection systems need to balance detection accuracy and real-time efficiency on resource-constrained platforms [12]. These difficulties are further amplified when the objects of interest occupy only a few pixels. Representative targets, including vehicles, ships, aircraft, and storage tanks, usually appear as small objects in large-scale aerial or satellite images. Consequently, small-object detection has become a critical and long-standing challenge in remote sensing object detection.

Small-object detection is inherently challenging due to the limited visual information contained in small objects and their

Received 29 June 2026; revised 23 August 2026; accepted 1 September 2026. Date of publication 3 September 2026; date of current version 18 September 2026. This work was supported in part by the Basic Research General Project of Liaoning Provincial Department of Education under Grant LJ212410142049, in part by the Joint Fund General Project of Liaoning Provincial Department of Science and Technology under Grant 2024-BSLH-206 and Grant 2024-MSLH-352, in part by the Science Foundation of Liaoning Province under Grant 2025-BS-0310, in part by the National Natural Science Foundation of China under Grant 62506240, and in part by the General Project of the Liaoning Provincial Department of Science and Technology Joint Fund under Grant 2023-MSLH-251. *(Corresponding author: Jian Liu.)*

The authors are with the School of Software, Shenyang University of Technology, Shenyang 110870, China (e-mail: yuanye@sut.edu.cn; lsl635@smail.sut.edu.cn; jianliu@sut.edu.cn; zsdiao@sut.edu.cn; tonggy@sut.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3730688

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

high susceptibility to confusion with background structures. In deep neural networks, successive downsampling operations further weaken or even erase small-object feature responses. Although high-resolution shallow features preserve rich spatial details, their semantic abstraction capability is limited and easily affected by background noise. In contrast, deep features provide stronger semantic representation but may lose fine-grained localization cues. Therefore, an effective detector for remote sensing small objects must achieve a proper balance between local detail preservation, semantic feature representation, and contextual modeling.

However, existing backbones primarily rely on local receptive fields and are unable to fully capture long-range contextual dependencies necessary for distinguishing small objects from complex backgrounds. Directly applying global modeling to all feature channels not only incurs high computational cost but may also disrupt critical fine-grained local features. FasterNet [13] alleviates part of these issues through efficient convolution and lightweight modules; nevertheless, it mainly focuses on local feature enhancement and model efficiency, and does not sufficiently coordinate local details and global context across different stages. These limitations highlight the need for a multiscale feature extraction mechanism that effectively balances local detail and global contextual information.

Moreover, multilevel feature fusion is another critical factor affecting small-object detection performance. Existing fusion strategies, such as simple addition or concatenation, are typically static and fail to explicitly model complementary relationships across different feature levels, which may result in weak small-object responses being overwhelmed by background noise. The modulation fusion module (MFM) [14] addresses this to some extent by adaptively fusing features from different levels through channel attention, enhancing feature robustness and discriminability. However, in remote sensing images, small-object detection requires not only discriminative channel selection but also precise spatial localization. Therefore, it is necessary to develop a collaborative fusion mechanism that simultaneously considers channel-wise complementarity and coordinate-aware spatial representation.

Query-based detectors still face certain limitations in remote sensing small-object detection. Most existing methods employ a fixed number of object queries for all images, without accounting for the actual object density. Many existing DETR-like approaches [15], [16], [17], [18], [19] rely on a predetermined number of object queries for all images, which simplifies implementation but can compromise recall in crowded scenes and result in unnecessary computations in sparse ones. In extreme cases, some aerial images contain more than 1500 tiny objects, far exceeding the typical query capacities of DETR (K = 100) [15] and Deformable-DETR (K = 300) [20]. To address these limitations, DDQ-DETR [21] increases the query set to K = 900 and employs a class-agnostic nonmaximum suppression (NMS) with manually defined IoU thresholds to remove redundant predictions. Despite these improvements, the method remains constrained by fixed-query counts and NMS parameters, limiting its recall in densely populated scenarios and its responsiveness to variable object densities. DQ-DETR [22] proposes a dynamic query mechanism using a classification-based counting module to allocate queries according to estimated object counts. While this strategy shows potential, it depends on manually tuned hyperparameters and introduces significant computational overhead due to the design of the counting module. These challenges highlight the urgent need for a density-guided dynamic query (DGDQ) mechanism, which adaptively allocates the number and spatial positions of detection queries based on a density map. Such a mechanism can reduce missed detections in densely populated regions and false positives in sparse areas, while avoiding manual parameter tuning and maintaining computational efficiency.

Based on the above observations, we propose SPEFormer, an end-to-end detection framework for remote sensing small objects. The core strategy of SPEFormer is to enhance small-object perception in a synergistic manner across three key stages: feature encoding, cross-level feature fusion, and query generation. Given an input image, SPEFormer first employs the selective global–local perception network (SGLPNet) to extract multiscale features. SGLPNet is designed to alleviate fine-detail degradation and background interference by combining conditional local convolution with deformable global attention [23] at different network stages. The extracted multilevel features are then fed into the adaptive synergistic attention fusion (ASAF) module, which adaptively integrates shallow spatial details and deep semantic cues through collaborative channel attention and coordinate-aware spatial attention [24]. This design strengthens discriminative small-object responses during feature fusion. Finally, a DGDQ mechanism predicts density-aware priors from encoder features and uses them to guide query allocation and selection. As a result, the decoder can reduce redundant queries in sparse scenes while improving recall in dense small-object regions. Through these designs, SPEFormer forms a unified perception-enhanced Transformer framework that progressively improves small-object detection from feature representation to query decoding. The main contributions of this article are summarized as follows.

1) We propose SGLPNet, a selective global–local perception backbone network for remote sensing small-object detection. Through a stage-adaptive channel partitioning strategy, SGLPNet integrates conditional convolution, deformable attention, and identity information preservation, achieving a favorable balance among detail preservation, contextual modeling, and computational efficiency.
2) We design an ASAF module for multilevel feature aggregation. This module jointly models collaborative channel dependencies and coordinate-aware spatial information, thereby enhancing small-object representations while suppressing background interference.
3) We introduce a DGDQ mechanism for Transformer-based detectors. A multiscale density extractor (MsDE) predicts object-density priors, which are further used for density-guided feature enhancement (DGFE), adaptive query allocation, and density-aware top-k query selection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

4) Extensive experiments on public remote sensing object detection datasets demonstrate the effectiveness of the proposed method. Comprehensive ablation studies and visual analyses further verify the contribution of each component to improving small-object detection performance.

II. Related Work

A. Backbone Networks for Remote Sensing Small-Object Detection

The backbone network is fundamental for extracting robust features in remote sensing small-object detection, directly affecting detail preservation, contextual modeling, and multilevel feature fusion. Early approaches commonly employed standard convolutional networks (e.g., ResNet [25]) as backbones, which effectively capture edges, textures, and local structures, but repeated downsampling causes small-object responses to decay rapidly in deep feature maps. To improve computational efficiency and feature representation, lightweight backbones such as GhostNet [26] leverage low-cost feature redundancy operations to enhance local information; however, they remain limited in preserving fine-grained small-object structures and distinguishing complex backgrounds. To overcome the inherent local receptive field limitations, vision Transformers [27] and large-kernel convolutions have been introduced to model long-range dependencies, although the complexity of self-attention scales quadratically with token number. More recently, state-space models exemplified by MobileMamba [28] reduce global modeling complexity to linear, offering a lightweight solution for long-range feature extraction. Nevertheless, existing backbones still face critical challenges. Fine-grained features in shallow layers are easily lost, deep global context may inadequately suppress complex backgrounds, and full-channel transformations under limited computation can blur small-object boundaries. Therefore, there is an urgent need for a flexible backbone capable of jointly modeling dynamic local textures and selective global dependencies to enhance small-object feature preservation and detection performance.

B. Multiscale Feature Fusion for Remote Sensing Small-Object Detection

Feature fusion is a critical step for integrating shallow spatial details and deep semantic context in small-object detection. Classical FPN [29] achieves multiscale feature aggregation through top-down pathways and lateral connections, while PANet [30] and BiFPN [31] further introduce bottom-up paths or weighted fusion strategies to enhance cross-level information interaction. However, most existing methods rely on simple addition, concatenation, or static weight assignment, making it difficult to adaptively regulate the contributions of different feature levels according to target distributions and background complexity. Consequently, weak tiny-object responses can be easily overwhelmed by background noise or high-level semantic information during feature fusion. To alleviate detail degradation, DPCF [32] balances high-resolution details and low-resolution context through channel grouping and learnable gating, effectively mitigating the dilution of small-object cues caused by fixed fusion strategies. Nevertheless, it mainly focuses on gated weighted fusion across scales, while its ability to model cross-level channel complementarity and precise spatial positional dependencies remains limited. Although channel and spatial attention mechanisms have been widely adopted to improve the discriminability of fused features, attention along a single dimension is insufficient to simultaneously preserve fine-grained structures, exploit cross-level semantic complementarity, and achieve accurate localization. Therefore, designing an adaptive fusion mechanism with both channel-synergistic modeling and spatial position-aware capability is essential for enhancing tiny-object representations in remote sensing images.

C. End-to-End Detectors for Remote Sensing Small-Object Detection

Object detection has gradually evolved from conventional CNN-based detectors to end-to-end Transformer-based detectors. Typical CNN-based detectors, such as Faster R-CNN [33], FCOS [34], and the YOLO series [35], [36], rely on convolutional feature extraction and dense prediction to perform object localization and classification, and have achieved remarkable progress in generic object detection. Among them, the YOLO series has been widely adopted in real-time detection scenarios due to its high inference efficiency. However, these methods generally still rely on post-processing operations such as NMS, and remain limited in small-object feature representation and long-range contextual modeling. This issue becomes more pronounced in remote sensing images, where small objects contain limited pixel information and their feature responses are easily weakened by successive downsampling in deep networks, thereby increasing the difficulty of detection in complex backgrounds.

The emergence of DETR provides a new end-to-end paradigm for object detection [15]. By using a fixed set of learnable object queries, DETR formulates object detection as a set prediction problem, eliminating hand-crafted components such as anchor design and NMS. Nevertheless, the original DETR suffers from slow convergence and high computational cost, which limits its application in efficient detection scenarios. To address these issues, subsequent methods such as Deformable-DETR [20] and DINO [37] improve detection performance and convergence efficiency by refining attention mechanisms, query design, and training strategies. Meanwhile, efficient DETR variants, such as RT-DETR [18], LW-DETR [38], and D-FINE [19], further promote the development of end-to-end detectors toward high-efficiency detection. In recent years, query-based Transformer detection frameworks have been increasingly explored for remote sensing object detection to cope with complex backgrounds, scale variations, and uneven object distributions. UAV-DETR [39] develops an efficient end-to-end detection framework for UAV imagery, whereas Freq-DETR [40] incorporates frequency-aware modeling for real-time small-object detection in UAV scenarios. Nevertheless, existing DETR-like methods still suffer from evident limitations when detecting small objects in remote sensing images. A key limitation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

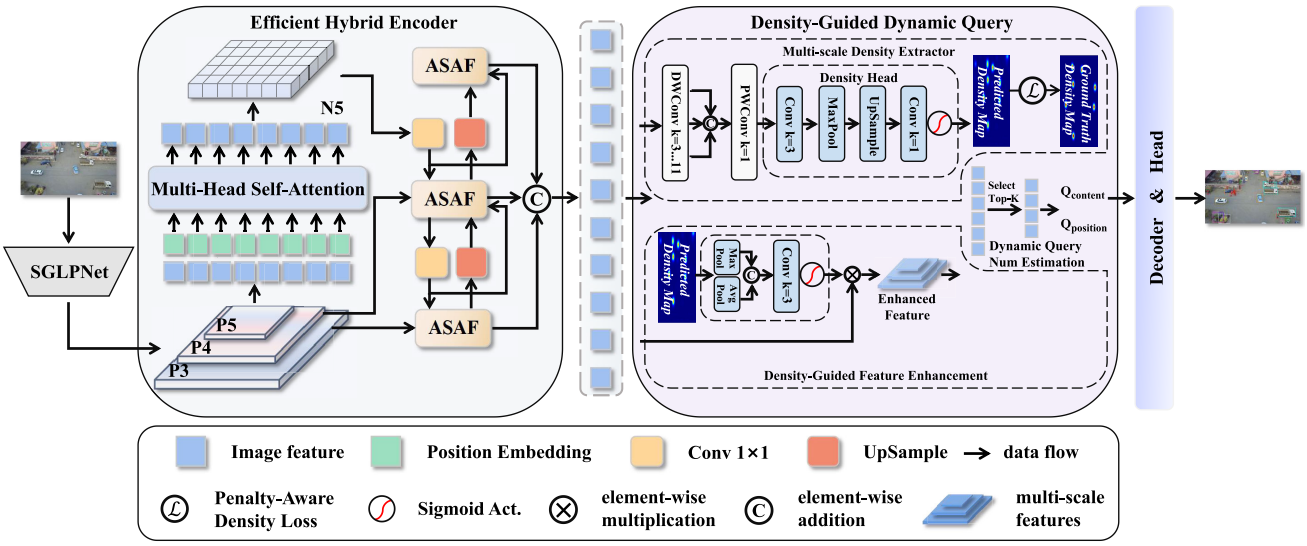


Fig. 1. Overall architecture of SPEFormer. The input image is first fed into the SGLPNet to extract multiscale features, where dynamic local textures of tiny objects are preserved while long-range global context is captured. The extracted features are then processed by the ASAF module, which suppresses complex background interference through cross-level channel collaboration and coordinate-aware spatial enhancement. Finally, the DGDQ module predicts an object-density map under the supervision of the PADL, and leverages the density prior to enhance target features and dynamically allocate the number and initial anchor positions of decoder object queries. The detection head then produces the final bounding-box predictions.

lies in the pronounced spatial heterogeneity and nonuniform distribution of remote sensing objects. For example, vehicles may be densely clustered in parking lots, whereas ships may be sparsely distributed over vast sea areas. Existing DETR-like detectors typically adopt a fixed number of object queries for all images, and this static query mechanism is difficult to adapt to drastic variations in object number and spatial density. In dense small-object regions, a fixed-query number may be insufficient to cover all potential objects, leading to missed detections. In large background regions, excessive redundant queries may introduce unnecessary background interactions and computational overhead, and may even increase the risk of false positives. Therefore, introducing object-density priors to adaptively regulate query allocation is of great importance for improving remote sensing small-object detection.

III. Method

A. Overview

As shown in Fig. 1, this study proposes SPEFormer, which is built upon D-FINE [19]. We enhance the baseline architecture with three key components: the SGLPNet for multiscale feature extraction, ASAF for generating more discriminative and localization-sensitive multiscale features, and DGDQ for adaptive query generation and selection.

B. Selective Global–Local Perception Network

We propose SGLPNet, as illustrated in Fig. 2. SGLPNet adopts a stage-adaptive channel partitioning strategy and designs two types of SGLPBlocks according to the feature requirements of different network stages: a shallow local perception form and a deep global–local perception form. In shallow stages, conditional convolution is applied to enhance only a subset of feature channels, while the remaining channels are preserved through identity mapping, thereby avoiding the disruption of fine-grained small-object cues. In deep stages, the input features are divided into global, local, and identity branches, which are responsible for global contextual modeling, dynamic local texture modeling, and stable information flow preservation, respectively. Finally, the features from different branches are concatenated and projected to generate enhanced feature representations.

1) Selective Global–Local Perception Block: To accommodate the different representation requirements of tiny objects at different network stages, we design a stage-adaptive SGLPBlock. The key idea is to avoid applying the same transformation to all feature channels. Instead, SGLPBlock selectively allocates different channel groups to local perception, global context modeling, and identity preservation according to the feature stage. In this way, the block can enhance discriminative information while reducing unnecessary disturbance to fragile small-object details.

In shallow stages, feature maps retain relatively high spatial resolution and contain rich edge, texture, and geometric cues. These fine-grained cues are crucial for localizing tiny objects. Therefore, the shallow SGLPBlock focuses on local detail preservation and adopts a partial channel processing strategy. Specifically, the input feature is divided into two parts: one part is fed into conditional convolution for dynamic local texture modeling, while the remaining part is directly preserved through an identity branch. This design enhances local patterns only on a subset of channels and keeps the original tiny-object responses unchanged in the remaining channels, thereby avoiding excessive feature transformation in high-resolution stages.

In deep stages, feature maps have stronger semantic representation but lower spatial resolution. At this stage, tiny-object recognition requires not only local texture information but also contextual cues from surrounding regions. Therefore, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

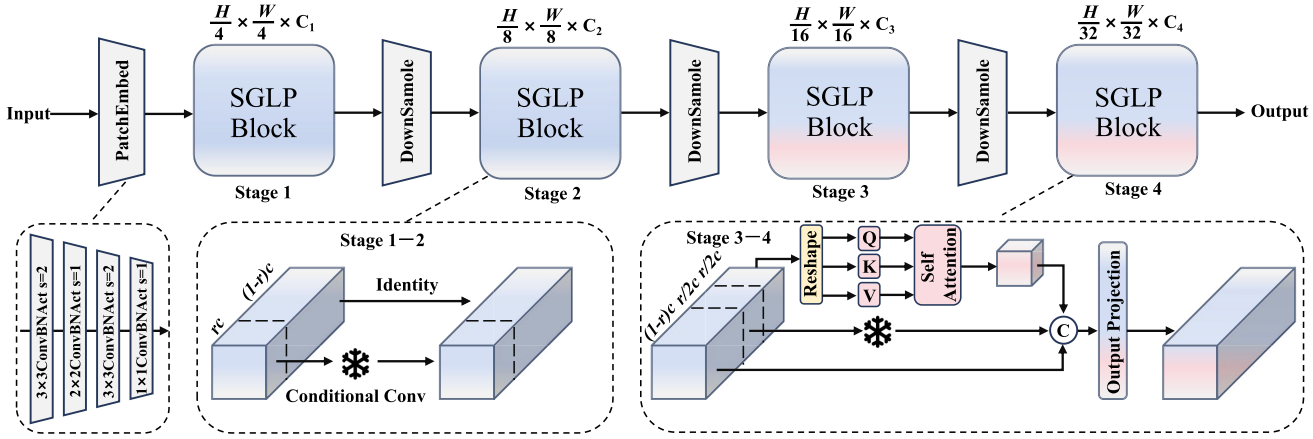


Fig. 2. Architecture of the selective global–local perception network. The proposed SGLPNet extracts multiscale features through stage-adaptive SGLPBlocks. Shallow SGLPBlocks (Stages 1 and 2) preserve fine-grained local details via partial conditional convolution, while deep SGLPBlocks (Stages 3 and 4) selectively integrate deformable global attention, conditional local modeling, and identity information flow to balance contextual representation and small-object detail preservation.

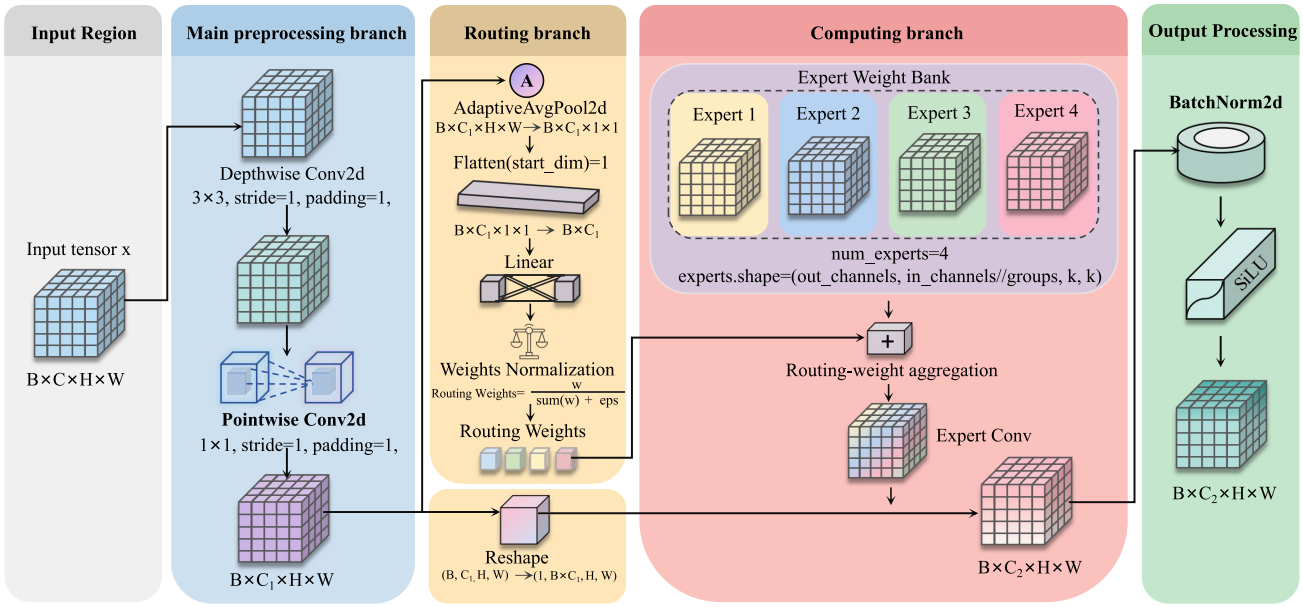


Fig. 3. Architecture of the Conditional Conv. The input feature is processed by depthwise and pointwise convolutions, and a routing mechanism generates input-dependent weights to combine multiple expert kernels. The resulting dynamic convolution kernel enables content-adaptive local perception for fine-grained small-object representation.

deep SGLPBlock further introduces a selective global–local structure. The input channels are divided into three branches: a global branch, a local branch, and an identity branch. The global branch adopts deformable attention [23] to capture long-range contextual dependencies in a sparse and efficient manner. The local branch employs conditional convolution to model input-adaptive local patterns. The identity branch preserves the original information flow and stabilizes feature propagation.

SGLPBlock provides a more flexible perception mechanism for remote sensing small-object detection. In shallow stages, it preserves fine-grained local details by combining conditional local enhancement with identity mapping. In deep stages, it integrates global context, adaptive local texture, and identity-preserved information in a unified block. This stage-adaptive design enables SGLPNet to maintain small-object details while introducing useful contextual information, achieving a better balance between representation capability, computational efficiency, and feature stability.

2) Conditional Convolution: For small objects, edge and texture features are highly susceptible to interference from complex backgrounds. To enhance the modeling capability for local textures and geometric details, we introduce conditional convolution into the SGLPBlock. Unlike standard convolution with fixed kernels, conditional convolution employs a routing mechanism to dynamically generate the combination weights of multiple expert kernels according to the current input features, thereby enabling input-adaptive local feature extraction. As illustrated in Fig. 3, given an input feature $X \in \mathbb{R}^{B\times C\times H\times W}$, depthwise convolution and pointwise convolution are first employed to perform local spatial modeling and channel interaction

$$X' = \mathrm{PWConv}\left(\mathrm{DWConv}\left(X\right)\right). \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

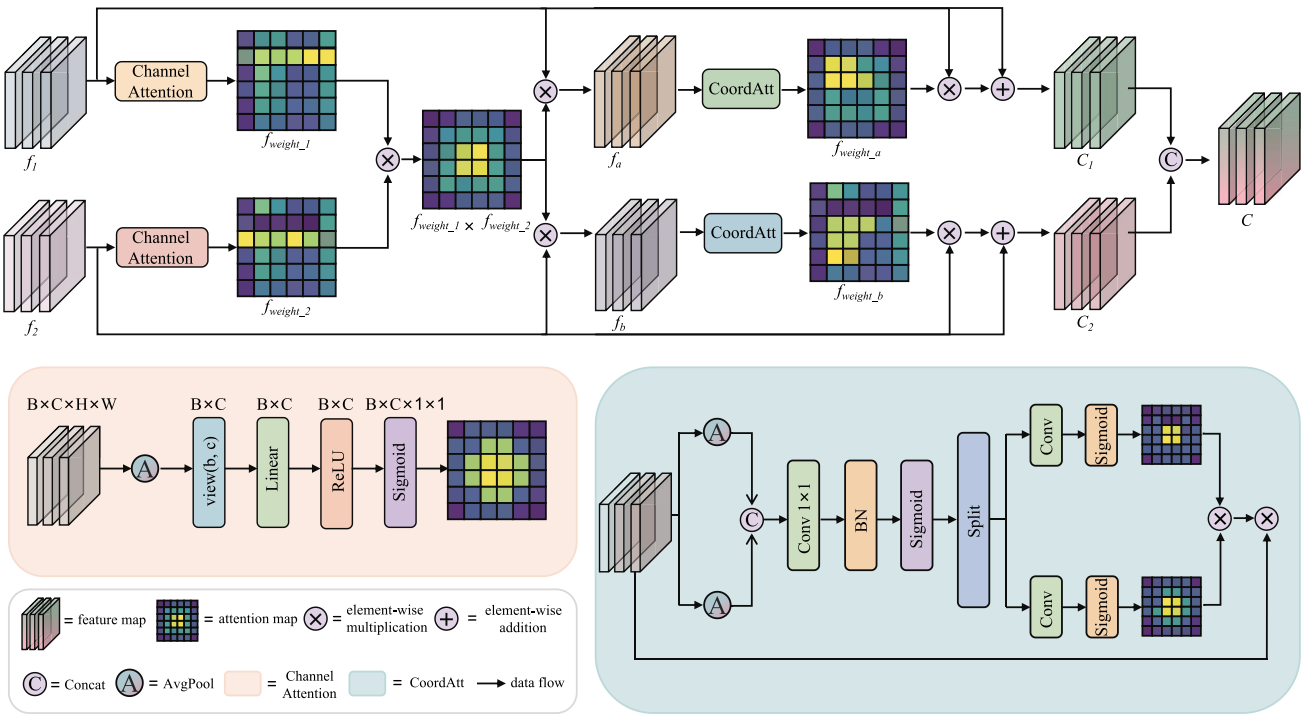


Fig. 4. Architecture of the ASAF. ASAF enhances cross-level feature fusion through collaborative channel attention and coordinate-aware spatial refinement. The collaborative channel attention highlights mutually informative channels between two input features, while coordinate attention improves spatial localization sensitivity.

Then, global average pooling is applied to X', followed by a fully connected layer to generate the routing weights for N expert convolution kernels

$$r = \sigma\left(\mathrm{FC}\left(\mathrm{GAP}\left(X'\right)\right)\right) \tag{2}$$

where $r \in \mathbb{R}^{B\times N}$ and $\sigma(\cdot)$ denotes the sigmoid function. To stabilize the contribution of different expert kernels, the routing weights are normalized as follows:

$$\hat{r}_n = \frac{r_n}{\sum_{j=1}^{N} r_j + \epsilon}. \tag{3}$$

Finally, the conditional convolution kernel is obtained by weighted aggregation of multiple expert kernels

$$W(X) = \sum_{n=1}^{N} \hat{r}_n W_n \tag{4}$$

and is then used to generate dynamic local features

$$Y = \mathrm{Conv}\left(X'; W(X)\right). \tag{5}$$

This design enables the network to adaptively select appropriate local perception patterns according to object morphology, background texture, and scale variation in different remote sensing scenes, thereby improving the representation capability of small objects.

C. Adaptive Synergistic Attention Fusion

We propose an ASAF module, as illustrated in Fig. 4, which enhances cross-level feature interaction in a target-aware manner.

Given two input features $F_1, F_2 \in \mathbb{R}^{B\times C\times H\times W}$, ASAF first performs channel attention on each feature independently. The channel descriptors are generated by global average pooling followed by a lightweight fully connected transformation, which produces channel-wise importance weights for the two input features. Instead of directly applying these two channel weights separately, ASAF multiplies them to obtain a collaborative channel attention map

$$A_c = A_1 \odot A_2 \tag{6}$$

where A_1 and A_2 denote the channel attention weights generated from F_1 and F_2, respectively, and $\odot$ denotes element-wise multiplication. The collaborative attention map A_c highlights the channels that are consistently informative across different feature levels, thereby suppressing redundant or inconsistent background responses. The two input features are then recalibrated by

$$R_1 = A_c \odot F_1, \quad R_2 = A_c \odot F_2. \tag{7}$$

After collaborative channel recalibration, ASAF further adopts coordinate attention [24] to enhance spatial position sensitivity. Coordinate attention encodes long-range dependencies along the horizontal and vertical directions while preserving positional information, which is beneficial for locating small objects in complex remote sensing scenes. In this work, coordinate attention is used as a spatial enhancement operation to refine the channel-recalibrated features R_1 and R_2, enabling the fusion module to emphasize potential target regions and suppress irrelevant background interference.

To avoid possible oversuppression of weak small-object responses, a residual enhancement strategy is further introduced. The coordinate-enhanced features are used to modulate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

the original input features, and the original features are added back to preserve complementary information

$$\hat{F}_1 = \text{CoordAtt}(R_1) \odot F_1 + F_1 \tag{8}$$

$$\hat{F}_2 = \text{CoordAtt}(R_2) \odot F_2 + F_2. \tag{9}$$

Finally, the enhanced features are concatenated along the channel dimension to generate the fused output

$$F_{\text{out}} = \text{Concat}\left(\hat{F}_1, \hat{F}_2\right). \tag{10}$$

Through this design, ASAF jointly considers cross-level channel synergy and coordinate-aware spatial enhancement. The collaborative channel attention emphasizes mutually informative channels between different feature levels, while coordinate attention improves the spatial localization sensitivity of small-object features. The residual fusion strategy further preserves original fine-grained cues and reduces the risk of attention-induced information loss. Compared with static fusion strategies, ASAF provides a more adaptive and discriminative feature aggregation mechanism for remote sensing small-object detection.

D. Density-Guided Dynamic Query

We propose a DGDQ mechanism, as illustrated in Fig. 1. This mechanism uses the density map as an explicit prior of object distribution and enhances the query generation process from three aspects. First, a density map is predicted to estimate potential object regions. Second, the density prior is used to enhance encoder memory features. Third, the number of object queries is dynamically adjusted according to the density response, and top-K query selection is jointly guided by classification confidence and density response.

1) Multiscale Density Extractor: To provide an explicit object-distribution prior for dynamic query generation, we design a MsDE to predict a density map from high-resolution encoder features. Since small objects are usually weak in semantic response but sensitive to local density distribution, the shallow encoder feature is used as the input to preserve more spatial details.

Given the shallow encoder feature F_s, the MsDE first employs multiple depthwise convolution branches with different kernel sizes to capture local density patterns under diverse receptive fields. The multiscale responses are then concatenated and projected by a pointwise convolution to obtain the density-aware feature representation. A lightweight density prediction head is further applied to generate a normalized density map through a sigmoid activation

$$D = \text{MsDE}(F_s) \tag{11}$$

where $D \in [0,1]^{B\times 1\times H_d\times W_d}$ denotes the predicted density map and MsDE$(\cdot)$ represents the proposed MsDE. The predicted density map provides a spatial prior indicating potential object regions and is subsequently used for DGFE and dynamic query generation.

To supervise the density estimation process, we generate a ground-truth density map for each image from the annotated bounding boxes using fixed Gaussian kernels. Specifically, for the bth image, let N_b denote the number of annotated objects. Each bounding box is represented in the center-based format (c_x, c_y, w, h). The center coordinates are first projected onto the spatial resolution of the predicted density map $H_d \times W_d$, and an impulse map $P_b \in \mathbb{R}^{H_d \times W_d}$ is constructed by assigning a unit response to each object center

$$P_b(y_i, x_i) = P_b(y_i, x_i) + 1 \tag{12}$$

where (x_i, y_i) denotes the discretized center location of the ith object on the density map. The impulse map is then smoothed by a Gaussian filter

$$\widetilde{D}_b^{gt} = \mathcal{G}_\rho(P_b) \tag{13}$$

where $\mathcal{G}_\rho(\cdot)$ denotes Gaussian filtering with a fixed standard deviation of $\rho = 15$ pixels. The Gaussian kernel is not adaptively adjusted according to the bounding-box size. In our implementation, zero padding is used at image boundaries. Since the Gaussian filter adopts a truncated support of 4ρ, the effective kernel size is

$$K = 2 \times \text{round}(4\rho) + 1. \tag{14}$$

Therefore, when $\rho = 15$, the effective kernel size is 121×121. After Gaussian smoothing, the density map is normalized to preserve the object count

$$D_b^{gt} = \widetilde{D}_b^{gt} \times \frac{N_b}{\sum_{x,y} \widetilde{D}_b^{gt}(x,y)}. \tag{15}$$

This count-preserving normalization ensures that the integral of the generated density map equals the number of annotated objects. For stable density supervision, both the predicted density map and the generated ground-truth density map are further peak-normalized before computing the density loss

$$\bar{D}_b = \frac{D_b}{\max(D_b) + \epsilon}, \quad \bar{D}_b^{gt} = \frac{D_b^{gt}}{\max\left(D_b^{gt}\right) + \epsilon} \tag{16}$$

where ϵ is a small constant for numerical stability.

A conventional mean squared error treats all pixels equally and does not explicitly distinguish between density underestimation and overestimation. However, in remote sensing small-object detection, underestimating density in object-dense regions may reduce small-object recall, while overestimating density in background regions may introduce redundant queries and false responses. Therefore, we introduce a penalty-aware density loss (PADL) to impose different penalties on these two cases. The penalty weight is defined as

$$\begin{aligned} W_b(x,y) = 1 &+ \lambda_r \bar{D}_b^{gt}(x,y)\, \mathbb{I}\left(\bar{D}_b(x,y) < \bar{D}_b^{gt}(x,y)\right) \\ &+ \lambda_p \bar{D}_b^{gt}(x,y)\, \mathbb{I}\left(\bar{D}_b(x,y) > \bar{D}_b^{gt}(x,y)\right) \end{aligned} \tag{17}$$

where $\mathbb{I}(\cdot)$ is the indicator function, λ_r denotes the recall-oriented penalty coefficient for density underestimation, and λ_p denotes the penalty coefficient for density overestimation in density-supported regions. Specifically, λ_r increases the penalty for underestimated density responses, encouraging the model to preserve more potential small-object regions. Since the overestimation penalty is modulated by $\bar{D}_b^{gt}(x,y)$, λ_p mainly acts on regions with nonzero ground-truth density generated by Gaussian kernels. For zero-density background pixels, the penalty weight reduces to 1, and false positive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

density responses are penalized by the basic squared error term.

Finally, the PADL is formulated as

$$\mathcal{L}_{den} = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{H_d W_d}\sum_{x,y}\ell_b(x,y) \tag{18}$$

$$\ell_b(x,y) = W_b(x,y)\left(\bar{D}_b(x,y) - \bar{D}_b^{gt}(x,y)\right)^2. \tag{19}$$

Through this penalty-aware supervision, the density extractor is encouraged to generate more reliable object-distribution priors, especially in dense small-object regions. This improves the quality of subsequent DGFE and query selection, enabling the detector to allocate query resources more effectively according to the spatial distribution of small objects.

2) Density-Guided Feature Enhancement: To encourage the encoder memory features to focus more on potential small-object regions, we design a DGFE module. First, the encoder memory $M \in \mathbb{R}^{B\times N\times C}$ is restored into multiscale feature maps according to the spatial shapes $(H_i, W_i)_{i=1}^{L}$

$$M_i \in \mathbb{R}^{B\times C\times H_i\times W_i}. \tag{20}$$

Then, spatial gating is performed on the multiscale density-related features generated by the density extractor. The spatial gate first compresses the channel dimension using channel-wise max pooling and average pooling

$$G_i = \left[\text{MaxPool}_c\left(F_i^d\right), \text{AvgPool}_c\left(F_i^d\right)\right]. \tag{21}$$

Subsequently, a convolution layer followed by a sigmoid function is used to generate the spatial weight

$$S_i = \sigma\left(\text{Conv}_{3\times3}\left(G_i\right)\right). \tag{22}$$

The generated spatial weight is then used to enhance the encoder feature at the corresponding level

$$\hat{M}_i = M_i \odot S_i. \tag{23}$$

Finally, all enhanced multiscale features are flattened and concatenated to reconstruct the enhanced encoder memory

$$\hat{M} = \left[\text{Flatten}\left(\hat{M}_1\right), \ldots, \text{Flatten}\left(\hat{M}_L\right)\right]. \tag{24}$$

Through DGFE, the encoder memory features can explicitly strengthen potential dense small-object regions and reduce the interference of background regions on subsequent decoder query selection.

3) Dynamic Query Number Estimation: Based on the predicted density map, we further design a dynamic query number estimation (DQNE) strategy to adapt the decoder queries to the object distribution of remote sensing images. Unlike conventional Transformer detectors that use a fixed number of object queries for all images, the proposed strategy dynamically estimates the query demand according to the density response. For each image, as shown in Fig. 5, the density response is accumulated and scaled by a query factor, and the resulting query number is constrained by predefined lower and upper bounds to ensure stable decoding. During mini-batch training, the maximum image-level query demand within the batch is adopted as the batch-level query number to maintain tensor consistency.

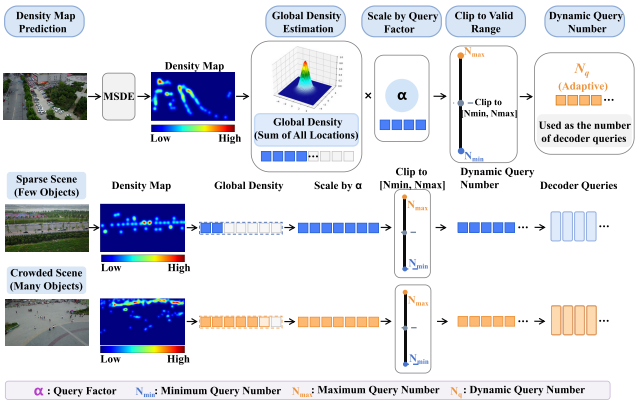


Fig. 5. Architecture of the DQNE.

In addition to query number allocation, the predicted density map is also used to guide top-K query selection. Specifically, the density map is first aligned with multilevel encoder tokens according to the feature strides. After a warm-up stage, the query selection score is computed by jointly considering the encoder classification confidence and the aligned density prior. In this manner, candidate locations with both high semantic confidence and strong density responses are more likely to be selected as decoder queries. This design alleviates the limitation of confidence-only query selection, where weak but true small-object responses may be suppressed by background regions.

The complete procedure of the proposed DQNE is summarized in Algorithm 1. For reproducibility, we provide the implementation details of DQNE. The query factor α, minimum query number $N_{\min}$, maximum query number $N_{\max}$, density temperature τ, score threshold θ, and density-guided starting iteration T_d are set to 6, 100, 1500, 10, 0.65, and 16 170, respectively. The density map is predicted by MsDE using multiscale depthwise kernels of 3, 5, 7, 9, and 11. To match the density prior with multilevel encoder tokens, we average-pool the predicted density map according to the feature stride of each encoder level, flatten the resulting maps, and concatenate them following the same token order as the encoder memory. Before T_d, top-Q candidates are selected using classification confidence only. After T_d, the aligned density prior is temperature-smoothed and max-normalized, and then multiplied by the classification confidence for density-guided query selection.

IV. Experiments

A. Datasets

1) VisDrone: VisDrone [41] is a large-scale drone-view object detection dataset containing 10 209 images, including 6471 images for training, 548 for validation, and 3190 for testing. It covers ten object categories and includes various object types, such as pedestrians, vehicles, and bicycles. The dataset also presents highly diverse object distributions, ranging from sparse scenarios to extremely crowded scenes.

2) DOTA-v1.0: DOTA-v1.0 [42] is a widely used benchmark for remote sensing object detection, characterized by high-resolution aerial images, dense object distributions, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 DQNE

Require: Encoder memory M, encoder logits P, anchors A, density map D, spatial shapes $\mathcal{S}$, feature strides $\mathcal{T}$, query factor α, query bounds $N_{\min}, N_{\max}$, temperature τ, threshold θ, iteration t, starting iteration T_d.

Ensure: Query content Q_c and reference boxes B_q.

Estimate the image-level query numbers from D and clamp them into $[N_{\min}, N_{\max}]$.
Set the batch-level query number Q as the maximum image-level query number in the mini-batch.
Apply the valid mask to M, and obtain refined encoder memory and logits P.
Align D with multi-level encoder tokens according to $\mathcal{S}$ and $\mathcal{T}$.
if $t < T_d$ **then**
Compute query scores using classification confidence.
else
Smooth and normalize the aligned density prior with τ.
    Compute query scores by combining classification confidence and the density prior.
    Suppress low-score candidates using threshold θ.
end if
Select top-Q candidate indices according to the query scores.
Gather the corresponding memory features and anchors.
Generate reference boxes B_q from selected anchors and predicted offsets.
Construct query content Q_c from learnable embeddings or selected memory features.
return Q_c, B_q.

large-scale variations. It contains 2806 large-scale images with resolutions ranging from 800×800–4000×4000 pixels, including 188 282 annotated instances from 15 categories. Notably, objects smaller than 32×32 pixels account for nearly 60% of all instances, making the dataset particularly challenging for small-object detection. Following common practice, the original images are cropped into 21 046 patches of 1024×1024 pixels with a stride of 824, resulting in 15 749 training patches and 5297 validation patches. Complex scenarios such as crowded harbors and aircraft clusters further evaluate the robustness of detectors under dense distributions, scale changes, and background interference.

3) DIOR: DIOR [43] is a challenging remote sensing object detection dataset designed for multiscale object detection under complex background conditions. It contains 23 463 remote sensing images with 190 288 annotated object instances. The dataset covers 20 major object categories, including buildings, vehicles, and swimming pools, representing diverse and typical remote sensing scenarios. The training, validation, and test sets contain 5862, 5,863, and 11 738 images, respectively.

B. Evaluation Metric

In accordance with the evaluation protocol described in [44], AP, AP_{50}, and AP_{75} are adopted as the primary quantitative measures for assessing detection performance on the VisDrone, DOTA-v1.0, and DIOR benchmarks. In addition, AP_s and AP_m are reported to analyze performance across different object scales, while FPS and $FLOPs$ are included to evaluate inference efficiency and computational complexity, respectively.

C. Implementation Details

All experiments are conducted on a single NVIDIA RTX 3090 GPU for both training and evaluation. The models are trained for 200 epochs on VisDrone, 100 epochs on DOTA-v1.0, and 160 epochs on DIOR. AdamW [65] is adopted as the optimizer with a weight decay of 0.0001. The batch size is set to 4, and the initial learning rate is set to 0.0002 for all datasets. All input images are uniformly resized to 640×640 pixels. To ensure fair and consistent comparisons, we follow the same data augmentation strategy as D-FINE. In addition, we provide two model variants, namely a small version and a large version, with progressively increasing parameter scales and computational complexity.

In the experimental tables, the asterisk * denotes the use of an additional $P2$ detection head with a stride of 4. The added detection head is introduced to exploit higher-resolution features and enhance dense small-object detection. This head adopts the same prediction-branch design as the original detection heads and is optimized using the same loss functions. No additional supervision, extra training data, special loss term, or different training strategy is introduced.

It should be noted that the * setting is consistently applied to both the baseline method and our method. Specifically, D-FINE* is obtained by adding the $P2$ detection head to D-FINE, while SPEFormer* is obtained by incorporating the proposed modules into D-FINE*. Similarly, SPEFormer is built upon the standard D-FINE. Therefore, the comparisons between D-FINE and SPEFormer, and between D-FINE* and SPEFormer*, constitute two fair and controlled comparisons under the same detection head configuration.

D. Comparative Results

1) Results on VisDrone: Table I reports the main experimental results on the VisDrone dataset. We compare the proposed SPEFormer with various representative detectors, including CNN-based methods and DETR-like methods. As shown in Table I, SPEFormer* achieves the best AP_{50} score of 54.7, outperforming existing state-of-the-art methods. It is worth noting that SPEFormer* is only slightly lower than SMWG-DETR in AP_m, with a marginal gap of 0.5 percentage points. Since SPEFormer is primarily designed to enhance small-object representations and density-aware query allocation, it achieves more significant gains on AP_s while maintaining competitive performance on medium objects. These results demonstrate that the proposed method provides stronger small-object detection capability on VisDrone and achieves superior performance over existing advanced DETR-like detectors.

2) Results on DOTA-v1.0: Table II presents the experimental results on the DOTA-v1.0 dataset. As shown in Table II, the proposed SPEFormer* achieves the best AP_{50}

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE VISDRONE DATASET. † DENOTES METHODS SPECIFICALLY DESIGNED FOR SMALL-OBJECT DETECTION, ∗ DENOTES THE VERSION WITH AN ADDITIONAL P_2 DETECTION HEAD WITH A STRIDE OF 4, AS DETAILED IN SECTION IV-C, AND BOLD INDICATES THE BEST RESULT

Method	Venue	Params	GFLOPs	AP	AP_{50}	AP_{75}	AP_s	AP_m
CNN-based methods								
Cascade-RCNN [45]	CVPR2018	69.3M	236.0	19.7	32.6	20.8	9.9	30.9
HIC-YOLOv5† [46]	ICRA2024	9.4M	31.2	26.0	44.3	28.5	14.8	37.6
YOLOv12 [47]	Arxiv2025	20.1M	67.2	26.4	43.6	28.1	16.4	36.3
YOLOv26 [48]	Arxiv2025	20.3M	67.9	26.3	43.2	27.6	16.3	36.0
FBRT-YOLO† [49]	AAAI2025	22.8M	185.8	30.1	48.4	31.7	21.8	38.1
DetectoRS w/RFLA† [50]	ECCV2022	134.8M	160.0	27.4	45.3	23.9	17.5	37.2
DNTR† [51]	TGRS2024	128.4M	298.0	29.2	48.3	30.1	18.9	42.3
DETR-like methods								
Salience-DETR† [52]	CVPR2024	55.9M	202.0	23.7	40.9	23.4	15.6	32.8
Deformable-DETR [20]	ICLR2021	40.0M	173.0	27.1	42.2	28.3	16.5	35.5
RT-DETR [18]	CVPR2024	42.0M	136.0	28.4	47.0	29.1	21.2	38.5
UAV-DETR† [39]	IROS2025	45.5M	247.1	32.4	51.9	33.6	24.1	42.9
SC-RTDETR† [53]	INFFU2024	44.3M	145.3	31.5	53.8	23.6	15.3	29.9
DQ-DETR† [22]	ECCV2024	58.7M	1762.4	33.2	52.9	33.8	24.9	43.8
DEIM [54]	CVPR2025	13.6M	37.4	25.4	42.2	25.8	17.7	34.9
DINO [37]	ICLR2023	47.6M	274.0	30.1	52.0	29.6	21.7	39.8
Position-DETR† [55]	TGRS2025	49.9M	184.0	25.3	43.3	25.4	17.1	34.4
SMWG-DETR† [56]	TGRS2026	49.9M	147.0	34.1	53.2	34.3	24.6	**44.7**
D-FINE [19]	ICLR2024	10.2M	24.8	26.7	44.6	27.2	18.7	36.5
D-FINE* [19]	ICLR2024	11.6M	72.9	32.1	51.6	33.3	24.0	42.8
SPEFormer (ours)	–	15.9M	54.1	28.5	47.8	28.6	21.5	38.3
SPEFormer* (ours)	–	18.4M	121.7	**35.0**	**54.7**	**35.6**	**26.7**	44.2

TABLE II

PERFORMANCE COMPARISON BETWEEN SPEFORMER AND OTHER DETECTORS ON THE DOTA-V1.0 DATASET. † DENOTES METHODS SPECIFICALLY DESIGNED FOR SMALL-OBJECT DETECTION, ∗ DENOTES THE VERSION WITH AN ADDITIONAL P_2 DETECTION HEAD WITH A STRIDE OF 4, AS DETAILED IN SECTION IV-C, AND BOLD VALUES INDICATE THE BEST RESULTS

Method	Venue	Params	GFLOPs	AP	AP_{50}	AP_s	AP_m	AP_l
CNN-based methods								
Faster-RCNN [33]	NeurIPS2015	52.1M	237.2	27.5	47.6	17.3	17.8	37.7
CenterNet [57]	ICCV2019	48.3M	112.6	20.8	45.2	13.5	13.9	36.3
EfficientDet [31]	CVPR2020	32.7M	593.4	21.1	46.3	14.8	13.7	36.9
YOLOv11 [58]	arXiv2024	2.6M	6.5	32.1	60.3	20.3	21.6	49.6
DETR-like methods								
Anchor-DETR [59]	AAAI2022	38.4M	176.5	32.9	60.8	25.7	22.3	50.2
Conditional-DETR [60]	ICCV2021	66.8M	229.2	26.4	51.3	15.6	13.5	40.8
Deformable-DETR [20]	ICLR2021	60.2M	228.2	32.7	62.4	25.7	22.4	51.5
SC-RTDETR† [53]	INFFU2024	39.2M	140.6	38.0	63.8	30.5	29.1	51.9
RT-DETR [18]	CVPR2024	41.6M	134.4	**39.8**	64.4	30.7	30.2	52.1
Salience-DETR† [52]	CVPR2024	55.9M	202.0	33.8	53.6	20.6	27.1	43.1
ARS-DETR† [61]	TGRS2024	43.7M	163.8	38.9	64.5	30.6	28.3	54.7
DINO [37]	ICLR2023	47.5M	284.0	36.2	62.2	18.8	39.4	46.0
D-FINE [19]	ICLR2024	10.2M	24.8	33.5	60.2	25.0	37.7	51.5
D-FINE* [19]	ICLR2024	11.6M	73.0	34.3	62.4	27.1	41.1	54.1
SPEFormer (ours)	–	15.9M	54.1	35.1	63.0	27.3	42.7	55.1
SPEFormer* (ours)	–	18.4M	121.8	37.3	**65.3**	**30.7**	**44.5**	**56.8**

performance. Compared with RT-DETR, SPEFormer* achieves a lower AP; however, it requires substantially fewer parameters and lower computational cost, indicating a more favorable accuracy–efficiency tradeoff.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE COMPARISON BETWEEN SPEFORMER AND OTHER DETECTORS ON THE DIOR DATASET. † DENOTES METHODS SPECIFICALLY DESIGNED FOR SMALL-OBJECT DETECTION, ∗ DENOTES THE VERSION WITH AN ADDITIONAL P_2 DETECTION HEAD WITH A STRIDE OF 4, AS DETAILED IN SECTION IV-C, AND BOLD VALUES INDICATE THE BEST RESULTS

Method	Venue	Params	GFLOPs	AP_{50}	AP_s	AP_m	AP_l
SuperYOLO† [62]	TGRS2023	7.7M	20.9	71.8	11.7	40.6	65.9
EfficientDet [31]	CVPR2020	32.7M	593.4	73.5	12.5	42.1	67.0
H-DINO [63]	CVPR2023	47.6M	299.0	85.7	20.4	50.0	78.0
Focus-DETR [64]	ICCV2023	49.3M	164.0	85.4	25.8	50.9	77.8
Anchor-DETR [59]	AAAI2022	76.3M	230.3	74.0	12.9	47.4	67.2
SMWG-DETR† [56]	TGRS2026	49.9M	147.0	65.1	10.7	36.2	51.5
SC-RTDETR† [53]	INFFU2024	79.6M	188.3	84.2	23.1	46.1	77.1
Salience-DETR† [52]	CVPR2024	55.9M	202.0	82.5	24.4	50.4	75.8
DINO [37]	ICLR2023	47.5M	284.0	79.7	21.7	50.7	73.6
ARS-DETR† [61]	TGRS2024	76.2M	167.4	81.2	19.6	46.7	75.0
Position-DETR† [55]	TGRS2025	49.9M	184.0	87.2	26.7	51.7	79.4
D-FINE [19]	ICLR2024	10.1M	24.9	82.0	20.9	44.6	75.9
D-FINE∗ [19]	ICLR2024	11.6M	73.0	85.7	24.7	48.3	79.7
SPEFormer (ours)	–	15.9M	54.1	84.1	23.0	46.7	77.1
SPEFormer∗ (ours)	–	18.4M	121.8	**87.7**	**26.9**	**51.7**	**80.5**

TABLE IV

ABLATION STUDY OF THE PROPOSED COMPONENTS ON THE VISDRONE DATASET

SGLPNet	ASAF	DGDQ	AP	AP_{50}	AP_{75}	AP_s	AP_m
×	×	×	26.7	44.6	27.2	18.7	36.5
✓	×	×	27.2	45.6	27.5	19.4	37.5
×	✓	×	27.2	45.7	27.7	19.3	37.8
×	×	✓	27.7	45.9	27.7	19.9	37.6
✓	✓	×	27.9	46.6	28.3	20.4	38.0
✓	×	✓	27.9	46.7	28.1	20.6	37.8
×	✓	✓	28.0	46.8	28.3	20.3	38.1
✓	✓	✓	**28.5**	**47.8**	**28.6**	**21.5**	**38.3**

3) Results on DIOR: Table III presents the experimental results on the DIOR dataset. As shown in Table III, SPEFormer∗ achieves an AP_{50} score of 87.7, outperforming the compared methods. Specifically, SPEFormer∗ improves AP_{50} by 2.0 percentage points over D-FINE∗ and by 0.5 percentage points over Position-DETR, demonstrating the effectiveness of the proposed method in complex remote sensing scenarios.

E. Ablation Study

The proposed method introduces three key components: the SGLPNet, the ASAF module, and the DGDQ mechanism. To evaluate the effectiveness of each component, we conduct a series of ablation studies using D-FINE as the DETR-like baseline model and systematically analyze the contribution of each module to the overall detection performance.

1) Main Ablation Experiment: Table IV presents the ablation results of the proposed components on the VisDrone dataset. It can be observed that each component contributes positively to the overall detection performance. Specifically, introducing SGLPNet, ASAF, and DGDQ individually improves AP_{50} from 44.6 to 45.6, 45.7, and 45.9, respectively. To further investigate the synergistic effects among the three components, we evaluate three pairwise combinations: SGLPNet+DGDQ, ASAF+DGDQ, and SGLPNet+ASAF. The results show that all three combinations outperform their corresponding single-component configurations. When SGLPNet, ASAF, and DGDQ are jointly integrated, the model achieves the best performance, with AP and AP_{50} reaching 28.5 and 47.8, respectively. These results demonstrate that SGLPNet, ASAF, and DGDQ provide complementary benefits for improving remote sensing small-object detection performance.

TABLE V

EXPERIMENTAL RESULTS OF DIFFERENT SGLPNET STRUCTURAL CONFIGURATIONS. BOLD INDICATES THE STRUCTURE ADOPTED IN THIS ARTICLE

SGLPBlock Stage 1–2	SGLPBlock Stage 3–4	Conv Type	Params	GFLOPs	AP_{50}	FPS
4	0	Conditional Conv	12.4M	26.8	45.1	92.1
0	4	Conditional Conv	12.6M	30.4	44.8	78.5
2	**2**	**Conditional Conv**	**12.5M**	**27.6**	**45.6**	**86.9**
2	2	Conv	12.2M	26.7	44.5	88.9

2) Ablation Study on the Architecture of SGLPNet: The ablation results in Table V validate the effectiveness of the SGLPNet architecture. When SGLPNet is constructed entirely with the SGLPBlock used in Stages 1 and 2, the model achieves a higher inference speed but relatively lower detection accuracy, indicating that local perception alone is insufficient for contextual modeling in deeper features. Conversely, using the SGLPBlock designed for Stages 3 and 4 across all stages increases computational cost and reduces FPS, suggesting that introducing global perception too early in shallow high-resolution features may disturb the fine-grained details of small objects. In contrast, the proposed stage-adaptive structure achieves the best AP_{50} while maintaining favorable computational efficiency, demonstrating its ability

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

SENSITIVITY ANALYSIS OF CHANNEL PARTITIONING RATIOS IN STAGES 3 AND 4 OF SGLPNET. THE ADOPTED SETTING IS HIGHLIGHTED IN BOLD

$r_g : r_l : r_i$	Params	GFLOPs	AP	AP_{50}	AP_{75}
0.125 : 0.125 : 0.75	12.1M	27.0	26.8	44.8	27.1
0.375 : 0.375 : 0.25	13.0M	28.5	27.1	45.2	27.2
0.25 : 0.25 : 0.50	**12.5M**	**27.6**	**27.2**	**45.6**	**27.5**

TABLE VII

SENSITIVITY ANALYSIS OF THE PENALTY COEFFICIENTS IN PADL. THE ADOPTED SETTING IS HIGHLIGHTED IN BOLD

λ_r	λ_p	AP	AP_{50}	AP_{75}	AP_s	AP_m
0.0	0.0	26.2	44.0	26.3	18.1	36.0
1.1	1.1	26.5	44.3	26.9	18.5	36.1
1.1	**1.3**	**27.7**	**45.9**	**27.7**	**19.9**	**37.6**
1.3	1.1	26.8	45.0	27.2	18.8	36.8
1.5	1.5	26.4	44.4	26.7	18.5	36.3

TABLE VIII

MSDE, DGFE, AND DQNE DENOTE MULTISCALE DENSITY EXTRACTOR, DENSITY-GUIDED FEATURE ENHANCEMENT, AND DYNAMIC QUERY NUMBER ESTIMATION, RESPECTIVELY

MsDE	DGFE	DQNE	AP	AP_{50}	AP_{75}	AP_s	AP_m
✓	✓		26.9	45.0	27.3	19.0	36.9
✓		✓	27.5	45.3	27.6	19.6	37.3
✓	✓	✓	**27.7**	**45.9**	**27.7**	**19.9**	**37.6**

TABLE IX

ABLATION STUDY ON THE QUERY FACTOR IN THE DGDQ MECHANISM. BOLD INDICATES THE SETTING ADOPTED IN THIS ARTICLE

Query Factor	**AP**	**AP_{50}**	**AP_{75}**	**AP_s**	**AP_m**
4	25.0	41.8	25.4	17.4	34.7
5	26.4	44.0	26.8	18.5	36.5
6	**27.7**	**45.9**	**27.7**	**19.9**	**37.6**
7	27.2	45.7	27.2	19.4	37.4
8	27.1	45.5	27.4	19.0	37.0

to better balance small-object detail preservation, contextual modeling, and computational cost. Moreover, replacing conditional convolution with standard convolution under the same structure decreases AP_{50} from 45.6 to 44.5, further confirming the effectiveness of conditional convolution for local feature modeling.

3) Effect of Channel Partitioning Ratios: In Stages 3 and 4 of SGLPNet, the input channels are divided into global, local, and identity branches. The global branch captures long-range contextual dependencies, the local branch models adaptive local patterns, and the identity branch preserves original feature information to reduce unnecessary feature disturbance. To analyze the influence of different channel partitioning ratios, we conduct an ablation study on $r_g : r_l : r_i$, where r_g, r_l, and r_i denote the channel ratios of the global, local, and identity branches, respectively. As shown in Table VI, increasing the identity ratio reduces the model complexity but leads to lower detection accuracy, indicating insufficient global–local feature transformation. Conversely, increasing the global and local ratios slightly improves the representation capacity but introduces higher computational cost. The adopted setting, 0.25: 0.25: 0.50, achieves the best tradeoff between accuracy and complexity, obtaining 27.2AP, 45.6AP_{50}, and 27.5AP_{75}.

4) Effect of Penalty Coefficients in PADL: PADL introduces two penalty coefficients, λ_r and λ_p. λ_r is the recall-oriented penalty coefficient for density underestimation, which encourages the model to preserve more potential small-object regions by increasing the penalty for underestimated density responses. λ_p controls the penalty for overestimated density responses in density-supported regions. Since the overestimation penalty is modulated by the ground-truth density map, λ_p mainly acts on regions with nonzero Gaussian density responses, while false positive responses in zero-density background regions are penalized by the basic squared error term. To analyze the influence of these two coefficients, we conduct a sensitivity study with different combinations of λ_r and λ_p. As shown in Table VII, removing the penalty-aware weighting by setting $\lambda_r = \lambda_p = 0$ yields 26.2 AP. Introducing moderate penalty coefficients improves the performance, and the adopted setting $\lambda_r = 1.1$, $\lambda_p = 1.3$ achieves the best results, with 27.7 AP, 45.9 AP_{50}, and 27.7 AP_{75}. When both coefficients are increased to 1.5, the AP drops to 26.4, indicating that overly strong penalty weights may disturb stable optimization. Therefore, we set $\lambda_r = 1.1$ and $\lambda_p = 1.3$ as the default configuration of PADL.

5) Ablation Study of Density-Guided Designs: To further analyze the source of the performance gain brought by the density-guided design, we conduct a detailed ablation study on three key components: MsDE, DGFE, and DQNE. Here, DQNE denotes the complete dynamic query process, including DQNE and density-guided Top-K selection. It should be noted that these components are not completely independent. MsDE provides the fundamental density prior and is required by both DGFE and DQNE. In addition, DQNE and density-guided Top-K selection are functionally coupled. The former estimates the number of queries according to the predicted density distribution, while the latter selects high-density spatial locations as query candidates. Therefore, we treat them as one complete DQNE process in the ablation study. As shown in Table VIII, the MsDE+DGFE setting achieves 26.9AP, 45.0AP_{50}, and 27.3AP_{75}, showing that DGFE improves density-aware feature representation. Compared with MsDE+DGFE, the MsDE+DQNE setting improves AP from 26.9 to 27.5, indicating that DQNE with density-guided Top-K selection contributes more significantly by adaptively allocating queries and selecting density-relevant spatial positions. When DGFE and DQNE are jointly used, the full MsDE+DGFE+DQNE setting achieves the best performance, with 27.7AP, 45.9AP_{50}, 27.7AP_{75}, 19.9AP_s, and 37.6AP_m. These results demonstrate that DGFE and DQNE are complementary, and their combination leads to the most effective density-guided detection framework.

6) Effect of Query Factor in DGDQ: To analyze the influence of the query factor in the DGDQ mechanism, we vary its value from 4 to 8 for ablation experiments, as reported in Table IX. It can be observed that the model performance gradually improves as the query factor increases from 4 to 6, indicating that an appropriate increase in the query factor can allocate more decoder queries to potential object regions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

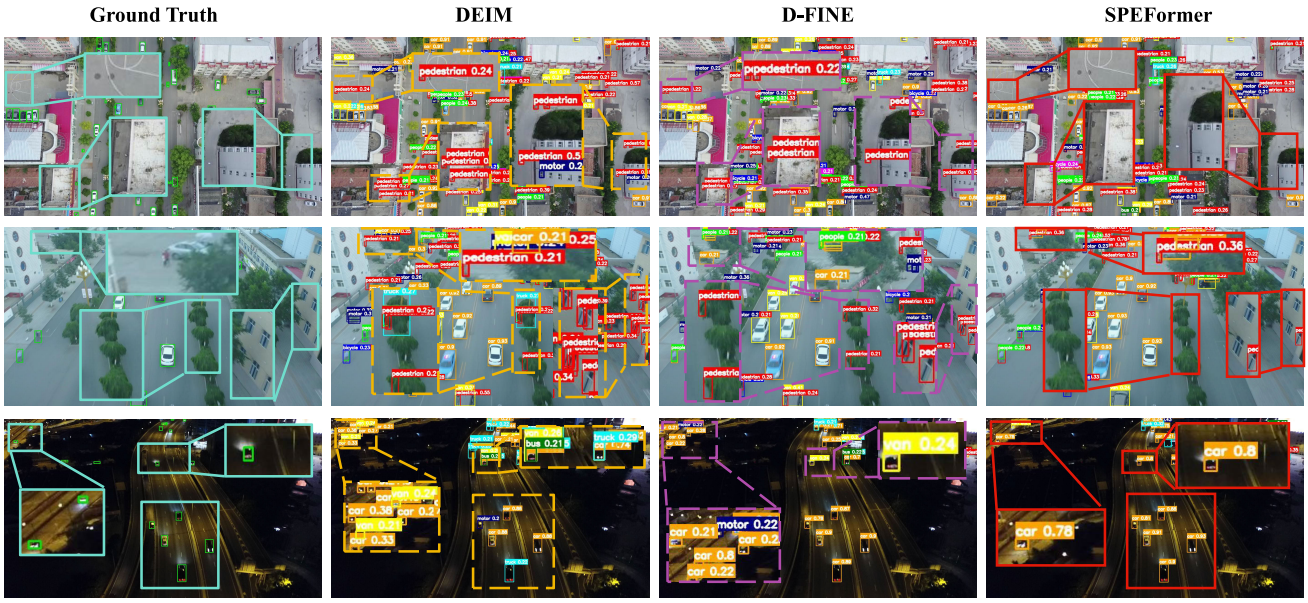


Fig. 6. Detection results on the VisDrone dataset. False detections and missed detections are highlighted with dashed bounding boxes.

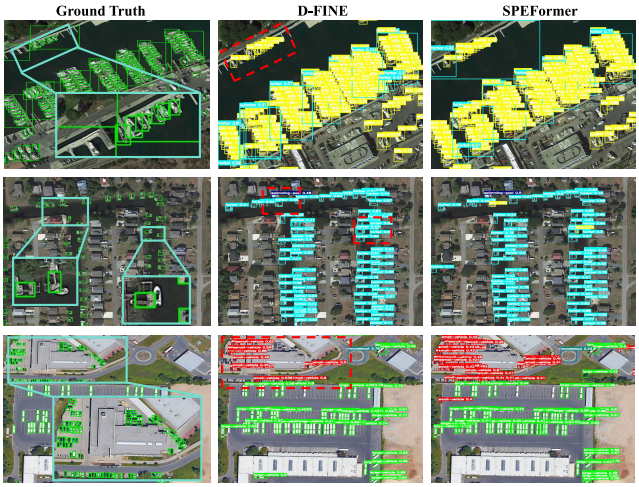


Fig. 7. Detection results on the DOTA-v1.0 dataset. False detections and missed detections are highlighted with dashed bounding boxes.

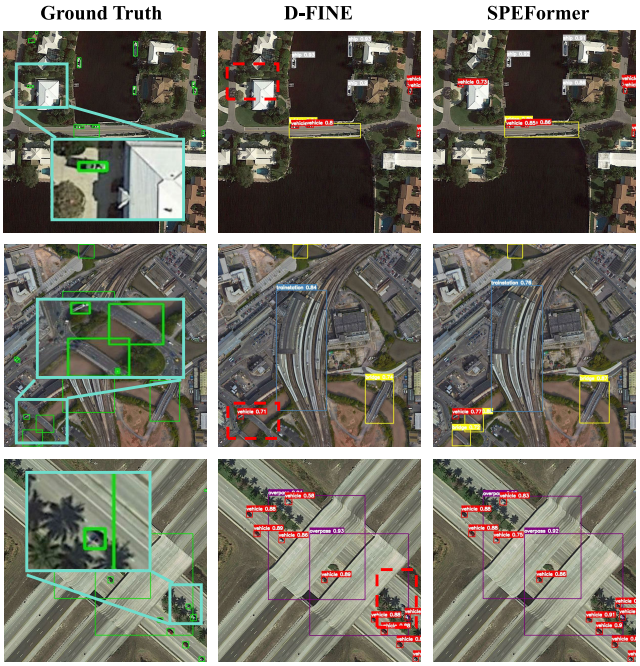


Fig. 8. Detection results on the DIOR dataset. False detections and missed detections are highlighted with dashed bounding boxes.

and thus improve small-object recall. The model achieves the best performance when the query factor is set to 6. However, further increasing the query factor to 7 or 8 leads to a slight performance degradation, suggesting that excessive queries may introduce redundant background candidates and reduce query effectiveness. Therefore, the query factor is set to 6 in the final model.

7) Query Number and Computation Analysis Under Different Object Densities: To further evaluate the adaptability of the proposed DGDQ mechanism, we compare SPEFormer with the fixed-query baseline D-FINE under different object-density intervals, as reported in Table X. The test images are divided into five groups according to the number of ground-truth object instances. During inference, the batch size is set to 1; therefore, the batch-level query number is equal to the image-level query number, i.e., $K_{batch} = K_{image}$. The reported average query number thus corresponds to the actual number of decoder queries used for each image. As shown in Table X, D-FINE adopts a fixed number of 300 object queries for all images, resulting in fixed decoder *FLOPs* of 2.0G. In contrast, SPEFormer dynamically adjusts the number of queries according to the predicted density prior. As the object density increases, the average query number of SPEFormer increases from 14 to 255, while the corresponding decoder *FLOPs* range from 94.5M to 1.7G, consistently remaining lower than those of the fixed-query baseline. These results indicate that DGDQ effectively reduces redundant decoder queries in sparse scenes while retaining sufficient query capacity in dense scenes. It should be noted that SPEFormer introduces additional density-related modules, including the MsDE and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

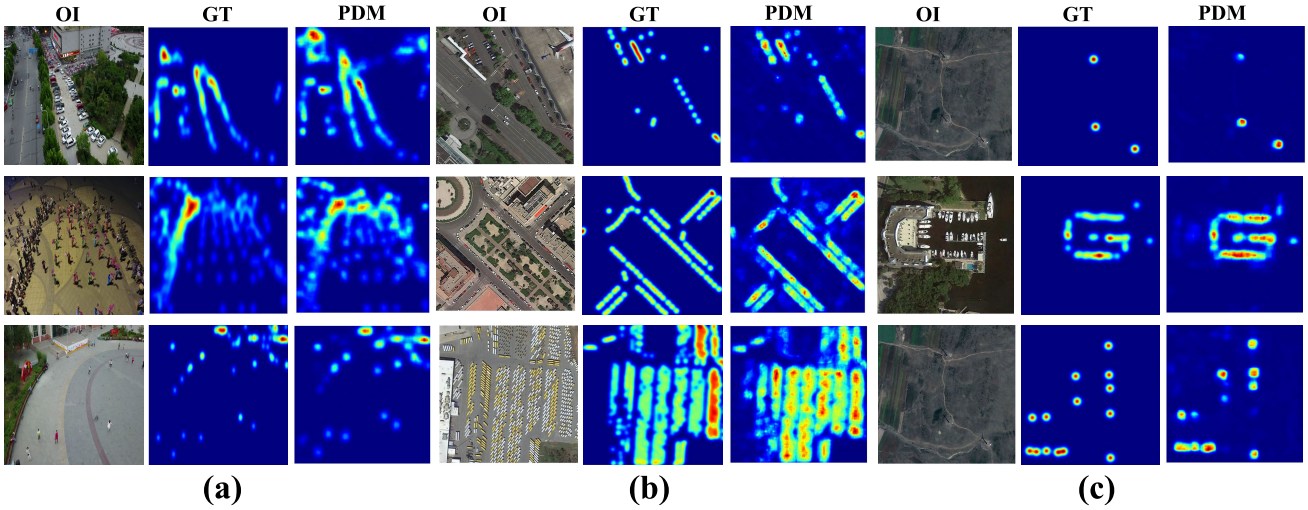


Fig. 9. Density map visualization results on three remote sensing datasets. (a)–(c) VisDrone, DOTA-v1.0, and DIOR, respectively. OI denotes the original image, GT denotes the ground-truth density map, and PDM denotes the predicted density map. Warmer colors indicate higher density responses.

TABLE X
QUERY NUMBER AND COMPUTATION ANALYSIS ACROSS DIFFERENT OBJECT-DENSITY RANGES ON VISDRONE

Model	Objects in Image	Avg. Query	Total FLOPs	Decoder FLOPs	FPS	AP	AP_{50}	AP_{75}
D-FINE	$N \leq 10$	300	24.8G	2.0G	10.8	23.9	38.3	24.7
	$10 < N \leq 50$				11.6	30.7	50.2	31.5
	$50 < N \leq 100$				11.6	27.0	45.2	27.1
	$100 < N \leq 200$				12.3	25.3	42.3	25.8
	$200 < N$				11.0	19.1	32.4	20.0
SPEFormer	$N \leq 10$	14	28.5G	94.5M	11.0	25.6	40.3	26.1
	$10 < N \leq 50$	62	28.9G	418.6M	10.9	31.0	50.7	31.6
	$50 < N \leq 100$	126	29.5G	851.0M	10.6	27.3	45.9	27.4
	$100 < N \leq 200$	210	30.3G	1.4G	11.4	25.7	43.6	26.0
	$200 < N$	255	30.7G	1.7G	10.7	21.7	39.4	21.8

DGFE. Consequently, its total *FLOPs* are higher than those of D-FINE. Nevertheless, SPEFormer maintains comparable practical inference speed to D-FINE across different object-density intervals, while consistently achieving better detection accuracy. This indicates that the additional density-related modules introduce only moderate practical inference overhead, while enabling more effective adaptive query allocation. In terms of detection performance, SPEFormer consistently outperforms D-FINE across all object-density intervals, with particularly notable improvements in extremely dense scenes. Overall, these results demonstrate that the primary advantage of DGDQ lies in adaptive query allocation and the reduction of decoder-side redundancy.

F. Visualization Analysis

Fig. 6 presents the visual comparison between the proposed method and other representative models under different detection scenarios on the VisDrone [41] dataset. It can be observed that, benefiting from the collaborative effects of SGLPNet, ASAF, and DGDQ, SPEFormer achieves superior detection performance in various complex scenes, especially in accurately localizing small objects and reducing false and missed detections.

Figs. 7 and 8 present visual comparisons between the proposed method and D-FINE [19] on the DOTA-v1.0 [42] and DIOR [43] datasets, respectively. Compared with D-FINE, SPEFormer exhibits superior detection performance in both dense and sparse scenes, achieving more accurate object localization while reducing false and missed detections.

Fig. 9 visualizes the density maps predicted by SPEFormer on three remote sensing datasets. It can be observed that the predicted density maps exhibit good spatial consistency with the ground-truth density maps and accurately highlight potential object regions. On VisDrone, PDM produces strong responses in road vehicle and crowded pedestrian regions. On DOTA-v1.0, PDM effectively characterizes the distribution patterns of densely arranged objects. On DIOR, even when the objects are relatively sparsely distributed, PDM still generates clear responses around a few target locations. These results indicate that the proposed density extraction branch can effectively learn object-distribution priors under different scenarios, providing reliable guidance for subsequent DGFE and dynamic query generation.

Fig. 10 presents the Grad-CAM visualizations of different methods in representative remote sensing small-object scenarios. It can be observed that DEIM and D-FINE can activate some target regions; however, their responses remain relatively scattered and contain redundant activations in complex background areas. In contrast, SPEFormer produces more concentrated and complete responses in road vehicle scenes, airport-related small-object regions, and densely distributed harbor ship scenes. In particular, in dense small-object scenarios, SPEFormer covers target instances more comprehensively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

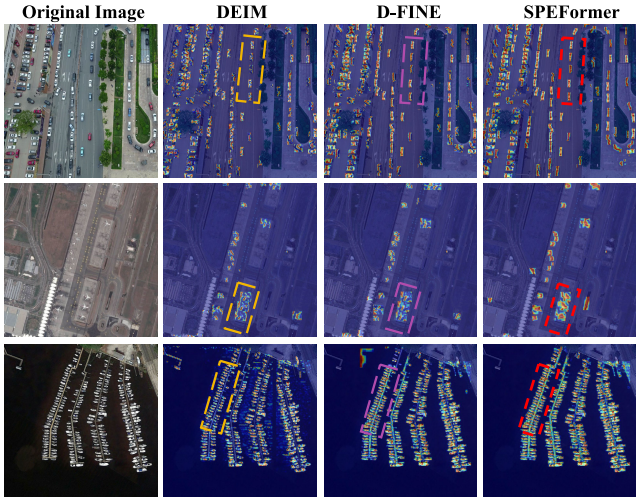


Fig. 10. Grad-CAM visualizations of different methods in representative remote sensing small-object scenarios. Warmer colors indicate stronger class-discriminative responses.

while maintaining relatively weak responses in large background regions. These visualization results indicate that the proposed method enhances discriminative feature representations for small objects and effectively improves the model's ability to focus on target-related regions under complex remote sensing backgrounds.

V. Conclusion

In this article, we propose SPEFormer, a synergistic perception-enhanced Transformer with density-guided dynamic queries for remote sensing small-object detection. The core strategy of SPEFormer is to jointly improve effective small-object feature representation and adaptive query allocation within an end-to-end detection framework. Specifically, SGLPNet enhances small-object representations by integrating dynamic local perception, global contextual modeling, and identity information preservation. ASAF further strengthens multiscale feature fusion through collaborative channel modeling and coordinate-aware spatial enhancement, preventing weak small-object responses from being overwhelmed by background interference. DGDQ exploits density priors to enhance encoder memory features and adaptively allocate decoder queries according to object distributions. Experimental results on the VisDrone, DOTA-v1.0, and DIOR datasets demonstrate the effectiveness of SPEFormer. In future work, we will explore more efficient query allocation strategies and advanced bounding-box representations to further improve the robustness and efficiency of remote sensing small-object detection.

References

[1] G. Wang et al., "Negative deterministic information-based multiple instance learning for weakly supervised object detection and segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6188–6202, Apr. 2025.

[2] W. Hua and Q. Chen, "A survey of small object detection based on deep learning in aerial images," *Artif. Intell. Rev.*, vol. 58, no. 6, p. 162, Mar. 2025.

[3] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[4] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[5] S. Chen, H. Dong, and W. Chen, "Scattering-term constrained nonnegative matrix factorization for hyperspectral unmixing," *J. Chongqing Univ. Posts Telecommun., Natural Sci. Edition*, vol. 33, no. 2, pp. 242–252, 2021.

[6] S. Chen and Y. Zhang, "Hyperspectral image compression based on prediction and vector quantization," *J. Chongqing Univ. Posts Telecommun., Natural Sci. Edition*, vol. 31, no. 5, pp. 688–693, 2019.

[7] Z. Lv, P. Zhang, P. Wang, W. Sun, M. Zhao, and R. Zhu, "Novel sample augmentation approach for improving classification performance with high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5643713.

[8] P. Wang et al., "CIRSM-Net: A cyclic registration network for SAR and optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610619.

[9] Z. He et al., "DADIGAN: A dual attention blocks-based disentangled iterative generative adversarial network for cloud and shadow removal on SAR and optical images," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103487.

[10] X. Zhang, Y. Feng, Z. Zhao, Y. Ren, and S. Mei, "Rethinking coarse-to-fine fully convolutional transformers for salient maritime ship detection in optical remote sensing imagery," *J. Remote Sens.*, vol. 6, p. 1038, Jan. 2026.

[11] X. Zhang, Y. Feng, N. Wang, G. Lu, and S. Mei, "Aerial person detection for search an d rescue: Survey and benchmarks," *J. Remote Sens.*, vol. 5, p. 0474, Jan. 2025.

[12] X. Zhang, Y. Feng, S. Zhang, N. Wang, G. Lu, and S. Mei, "Robust aerial person detection with lightweight distillation network for edge deployment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5630616.

[13] J. Chen et al., "Run, Don't Walk: Chasing higher FLOPS for faster neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12021–12031.

[14] Y. Zhang, S. Zhou, and H. Li, "Depth information assisted collaborative mutual promotion network for single image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2846–2855.

[15] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2020, pp. 213–229.

[16] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 13619–13627.

[17] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," 2022, *arXiv:2201.12329*.

[18] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.

[19] Y. Peng, H. Li, P. Wu, Y. Zhang, X. Sun, and F. Wu, "D-FINE: Redefine regression task of DETRs as fine-grained distribution refinement," in *Proc. Int. Conf. Learn. Represent.*, vol. 2025, 2025, pp. 44015–44031.

[20] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[21] S. Zhang et al., "Dense distinct query for end-to-end object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7329–7338.

[22] Y.-X. Huang, H.-I. Liu, H.-H. Shuai, and W.-H. Cheng, "DQ-DETR: DETR with dynamic query for tiny object detection," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2024, pp. 290–305.

[23] Z. Xia, X. Pan, S. Song, L. E. Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4794–4803.

[24] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13713–13722.

[25] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.

[26] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 1580–1589.

[27] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[28] H. He et al., "MobileMamba: Lightweight multi-receptive visual mamba network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 4497–4507.

[29] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[30] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[31] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[32] W. Xu et al., "SAMamba: Adaptive state space modeling with hierarchical vision for infrared small target detection," *Inf. Fusion*, vol. 124, Dec. 2025, Art. no. 103338.

[33] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[34] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[35] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[36] C. Wang et al., "Gold-YOLO: Efficient object detector via gather-and-distribute mechanism," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 51094–51112.

[37] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[38] Q. Chen et al., "LW-DETR: A transformer replacement to YOLO for real-time detection," 2024, *arXiv:2406.03459*.

[39] H. Zhang, H. Zhang, K. Liu, Z. Gan, and G.-N. Zhu, "UAV-DETR: Efficient end-to-end object detection for unmanned aerial vehicle imagery," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2025, pp. 15143–15149.

[40] J. Chen, N. Liu, H. Sun, and Y. Wang, "Freq-DETR: Frequency-aware transformer for real-time small object detection in unmanned aerial vehicle imagery," *Expert Syst. Appl.*, vol. 298, Mar. 2026, Art. no. 129710.

[41] D. Du et al., "VisDrone-DET2019: The vision meets drone object detection in image challenge results," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 213–226.

[42] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[43] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[44] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.

[45] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high Quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[46] S. Tang, S. Zhang, and Y. Fang, "HIC-YOLOv5: Improved YOLOv5 for small object detection," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2024, pp. 6614–6619.

[47] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 38, 2025, pp. 87151–87175.

[48] R. Sapkota, R. H. Cheppally, A. Sharda, and M. Karkee, "YOLO26: Key architectural enhancements and performance benchmarking for real-time object detection," 2025, *arXiv:2509.25164*.

[49] Y. Xiao, T. Xu, Y. Xin, and J. Li, "FBRT-YOLO: Faster and better for real-time aerial image detection," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 8, pp. 8673–8681.

[50] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 526–543.

[51] H.-I. Liu, Y.-W. Tseng, K.-C. Chang, P.-J. Wang, H.-H. Shuai, and W.-H. Cheng, "A DeNoising FPN with transformer R-CNN for tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704415.

[52] X. Hou, M. Liu, S. Zhang, P. Wei, and B. Chen, "Salience DETR: Enhancing detection transformer with hierarchical salience filtering refinement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17574–17583.

[53] H. Feng et al., "Security of target recognition for UAV forestry remote sensing based on multi-source data fusion transformer framework," *Inf. Fusion*, vol. 112, Dec. 2024, Art. no. 102555.

[54] S. Huang, Z. Lu, X. Cun, Y. Yu, X. Zhou, and X. Shen, "DEIM: DETR with improved matching for fast convergence," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 15162–15171.

[55] W. Zhao, X. Deng, F. Gao, and C. Liu, "Position-DETR: Step-by-step position-guided small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5641214.

[56] M. Chen, W. Zhao, N. Li, D. Li, R. Zhang, and J. Xu, "SMWG-DETR: DETR enhanced by Fourier spectral modulation and wavelet-guided fusion for tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5605117.

[57] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[58] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.

[59] Y. Wang, X. Zhang, T. Yang, and J. Sun, "Anchor DETR: Query design for transformer-based detector," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 3, Jun. 2022, pp. 2567–2575.

[60] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3651–3660.

[61] Y. Zeng, Y. Chen, X. Yang, Q. Li, and J. Yan, "ARS-DETR: Aspect ratio-sensitive detection transformer for aerial oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610315.

[62] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415.

[63] D. Jia et al., "DETRs with hybrid matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19702–19712.

[64] D. Zheng, W. Dong, H. Hu, X. Chen, and Y. Wang, "Less is more: Focus attention for efficient DETR," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6674–6683.

[65] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:39:29 UTC from IEEE Xplore. Restrictions apply.